Schell Carl Scivally

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My tl;dr

- I'm passionate about
    - functional programming
    - real-time rendering
    - music and sound synthesis
- I'm "T-shaped" engineer (a generalist with a few deep interests):
    - GPU Programming
    - Web Dev (APIs and user interfaces)
    - Programming Languages
    - AI Inference
- I feel equally comfortable working on the frontend or backend of a project.
- I love a good optimization problem.
- My preferred programming languages are Rust, Haskell, Objective-C, and JavaScript.

At a high level, this has been my career, chronologically:

1. Games
2. Agency work and retail
3. AI
4. A little crypto (zcash)

And besides working in software I have:

- Been an expert witness in a Texas courtroom
- Played in a couple rock bands in California
- Toured the US as a merch boy for the Warped Tour
- Driven a forklift in a chemical plant
- Bailed hay in three states
- Danced ballet in New Zealand

In my free time I like to do handy work, play games with my two kids and make "altered art".

This resume lives permanently at https://zyghost.com/resume.html, and you can download this resume as a PDF.

🏠 Current Location

I am a US citizen living in Nelson, New Zealand 🌴🇳🇿.

🔗 Links

- GitHub - https://github.com/schell
- LinkedIn - https://linkedin.com/in/schellscivally

Open Source Work

- I maintain many open source libraries as my current day job, working on the Zcash cryptocurrency protocol at ZODL.
- I am a two-time NLNet grantee, working on a cross-platform GPU-driven rendering ecosystem that runs on WebGPU.
    - 2024 Renderling
    - 2025 Renderling Ecosystem.
- Authored wgsl-rs, a proc-macro that transpiles Rust to WGSL so shaders can be unit-tested on the CPU and run isomorphically on the GPU, with automatic `wgpu` runtime linkage.
- Authored renderling, a 3d physically-based renderer with easy resource management.
- Authored crabslab, a slab allocator for working with GPUs in Rust.
- I was a maintainer of Rust-GPU, which is a Rust compiler that generates SPIR-V shaders.
- I added support for atomic operations to `wgpu`'s transpiler SPIR-V frontend.
- I contributed RenderLayers to the popular Bevy game engine.
- I authored typograffiti, a text renderer written in Haskell.
    - I am the initial author but have since moved on to `renderling`.
    - I handed off maintainership of `typograffiti` to Adrian Cochrane (**alcinnz** on github) who himself is a NLNet grantee.

Day Job Experience

ZODL Inc. (formerly Electric Coin Company) Jul 2025 - Present

Core Engineer

https://zodl.com/

Working on the Zcash protocol and core libraries.

Rust

Narrative.so Feb 2021 - Jul 2025 (4 years 5 months)

Senior Rust Software Engineer

https://narrative.so/

Reduced the time to deliver new AI models from R&D to production by 50%.

Integrated batch processing of ONNX models into AI inference systems in Rust.

Optimized a cross-platform consumer application that performs AI inference on the edge, including

- performance profiling and optimization
- shipping new AI models and features
- refactored build systems to use Rust `xtask`

Authored a realtime image editing application in Rust using `wgpu`, `WGSL` and `WASM`.

Rust, Python, Lua, C, C++, JavaScript, Typescript

Formation.ai Oct 2016 - Feb 2021 (4 years 5 months)

Senior Software Engineer

~~https://formation.ai/~~ (acquired by Boston Consulting Group in 2022)

Full stack engineering in Haskell using `reflex`. Rust, Go, AWS.

- single-handedly authored, deployed, and maintained a cloud-based template language interpreter in Rust that ran 24 million+ Starbucks req/day for 6 months - bug free!
- built a fullstack application in Haskell to author Starbucks' Star Program offers.

Formation.ai was acquired by BCG and was formerly Takt.com.

Haskell, Javascript, Rust

Y Syndeca.com Apr 2010 - Oct 2016 (6 years 7 months)

Senior Developer

https://syndeca.com/

Full stack engineering in Objective-C, Flash, Javascript and Haskell.

- single-handedly authored, deployed and maintained SDK and applications for clients including:
    - Nordstrom
    - Home Depot
    - Lowes
    - Bass Pro
    - Sears
- internal productivity tools in Haskell

Objective-C, JavaScript, Haskell

🤝 Freelancing 2005 - 2012

Flash game development, interactive displays, website design and implementation.

Clients include:

- Viddyou.com 2009 (acquired by Motionbox in 2013)
    - authored Viddyou's HD streaming video player
- Opsworks, Inc.
    - authored the Larchmont Charter School auction and payment system
- Synapse Group, Inc. (digital agency)
    - completed many projects for Fortune-500 clients like:
        - IBM
        - 3M
        - Sears
        - Lowes
        - Nordstrom

- Sonoma State University

NASA Education and Public Outreach Sep 2003 - Apr 2008 (4 years 9 months)

https://en.wikipedia.org/wiki/NASA_Education_and_Public_Outreach_Group

Flash game development.

- Designed and authored the “Solar Supernova” Space Mysteries game
- Authored, deployed and maintained many of the program’s various websites

Honors and Awards

Cultural Award - We’re Daring

Narrative.so Feb 2025

For excellent work planning and orchestrating the delivery of new AI models into production.

> We act decisively to make brave and bold choices.

Cultural Award - Embrace Creative Freedom

Narrative.so Nov 2023

For shipping consistenly and quickly, with an eye for creative solutions.

> We walk in artists’ shoes and embrace creative freedom. We’re curious, passionate and hunt for the deeper meaning.

Cultural Award - Seek Truth

Narrative.so Aug 2021

For work on integrating TVM tuning into AI model pipelines and revamping build systems.

> We seek out our strengths and weaknesses and accept them – at all times being authentically true to ourselves, and each other.

Mcquillen Summer Research Award

Sonoma State University Jun 2005

Selected to refurbish a thin film sputter coating machine and work on various high vacuum projects in the physics department.